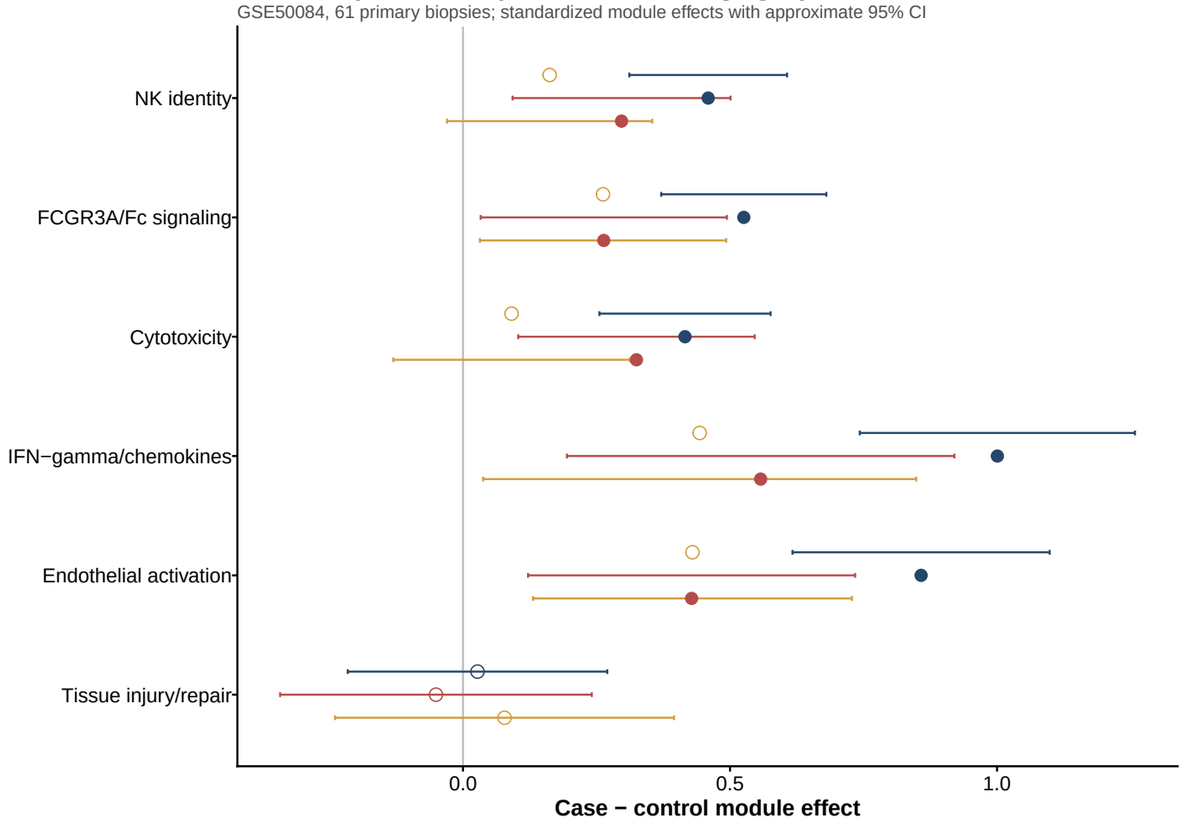
A DSA exposure is separated from damaging rejection

GSE50084, 61 primary biopsies; standardized module effects with approximate 95% CI

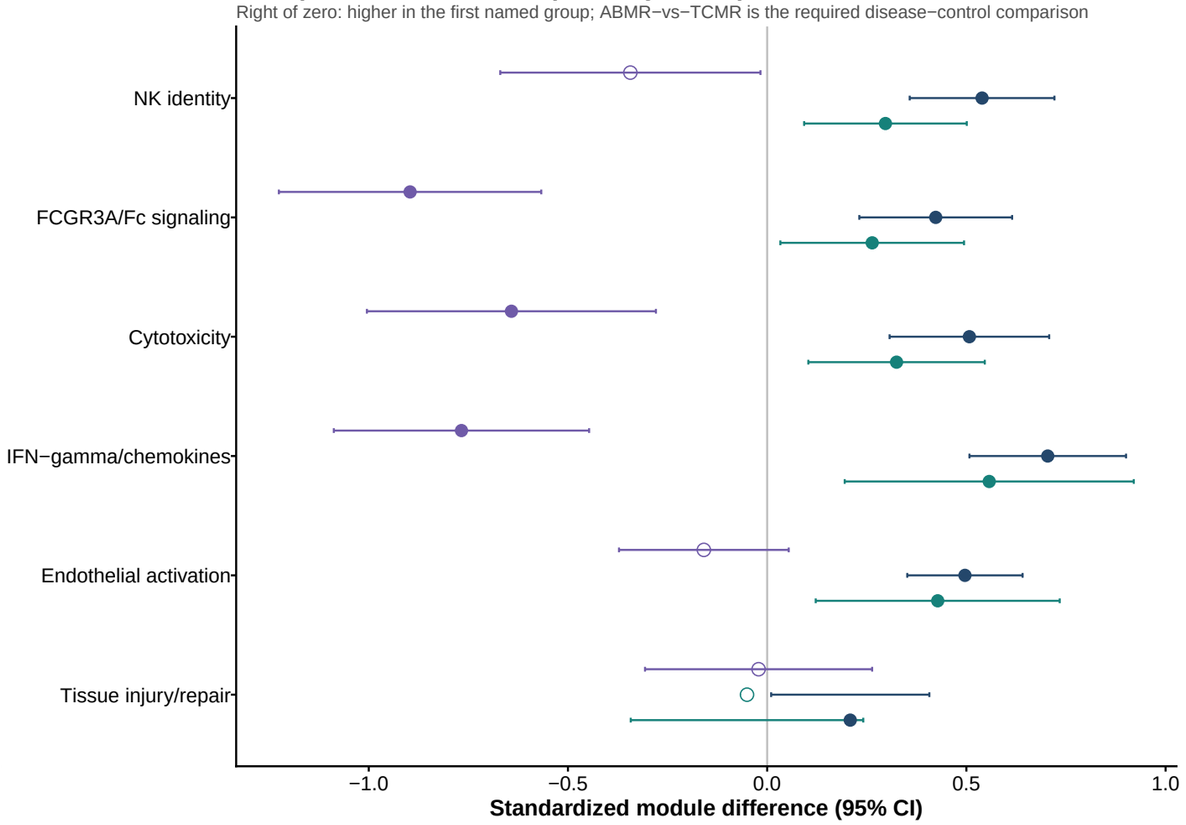


Estimand DSA exposure without rejection · Rejection among DSA+ grafts · DSA+ rejection vs DSA− control

Evidence FDR < 0.05 · FDR >= 0.05

B Replication and ABMR specificity are separate tests

Right of zero: higher in the first named group; ABMR−vs−TCMR is the required disease−control comparison

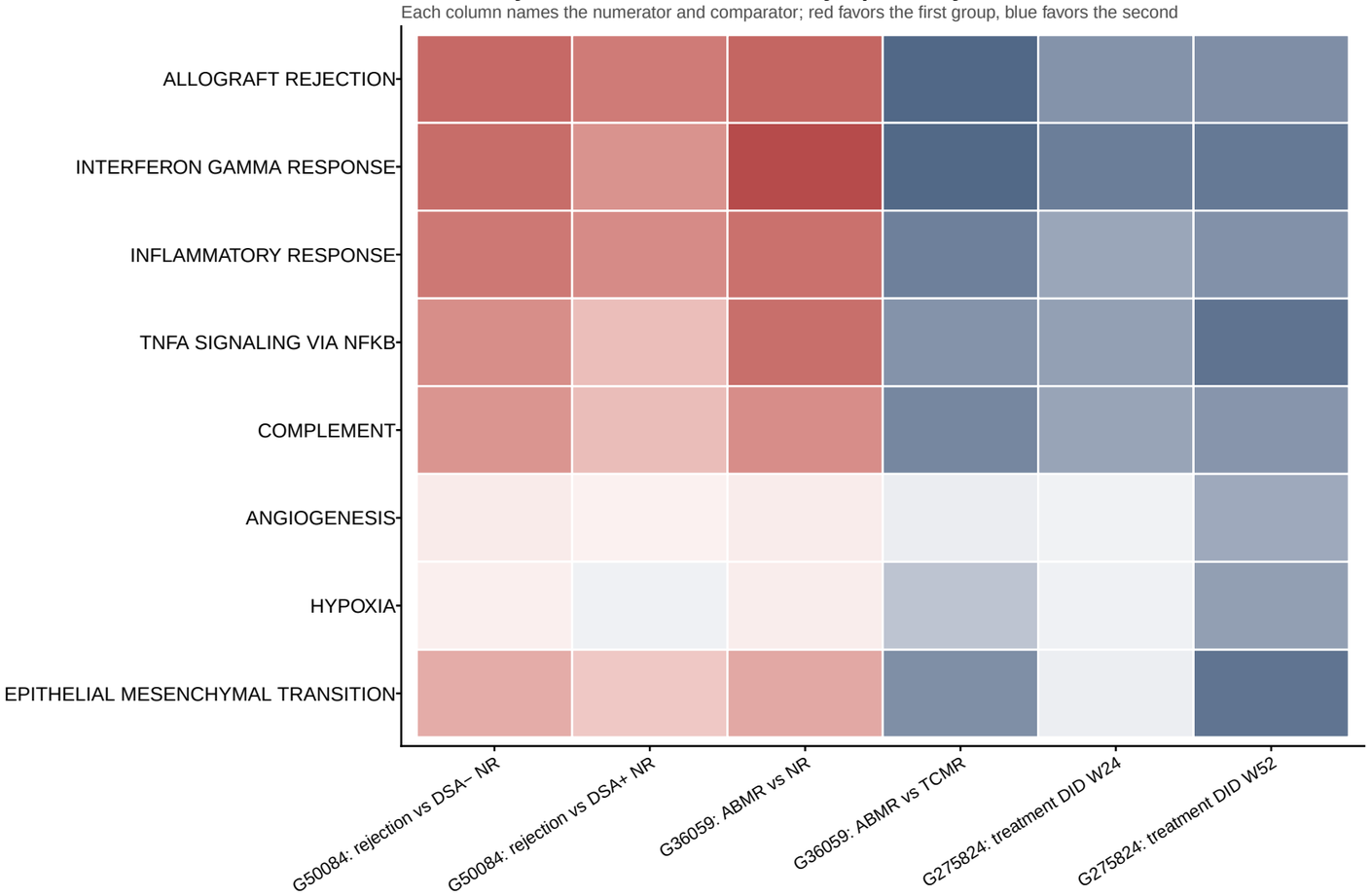


Comparison Discovery: DSA+ rejection vs DSA+ no rejection · Validation: ABMR vs non−rejecting · Specificity: ABMR vs TCMR

Evidence FDR

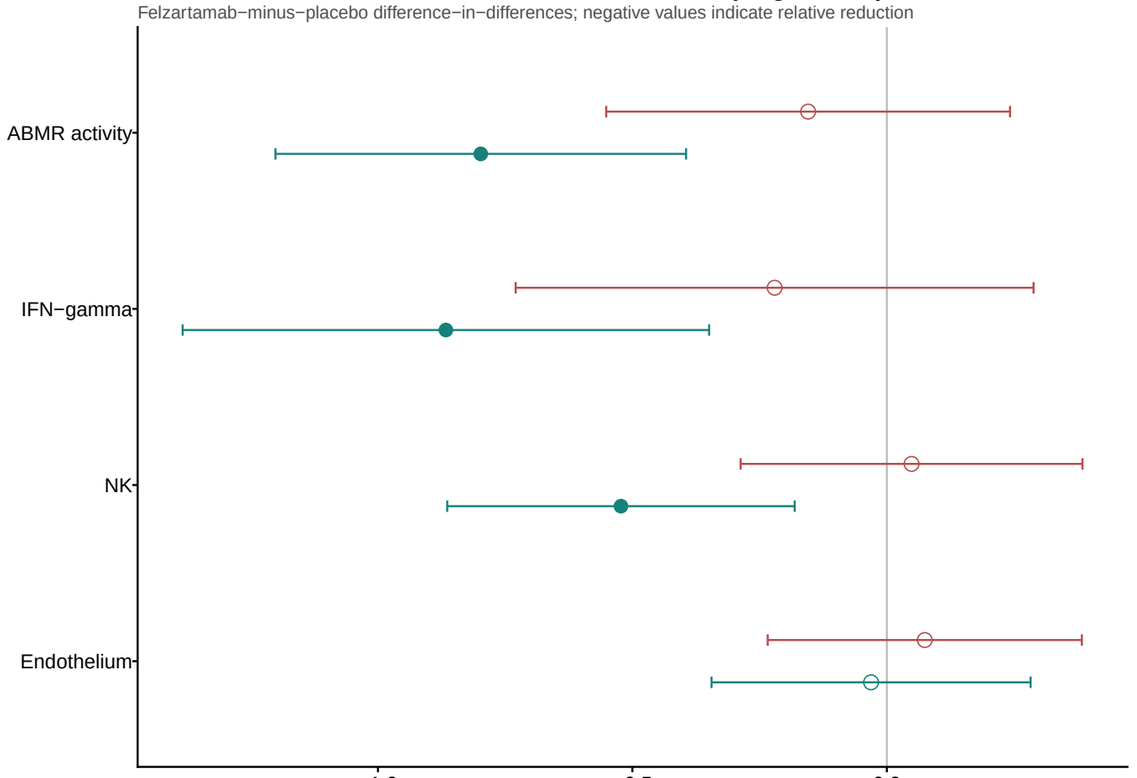
C Pathway direction across discovery, specificity and treatment

Each column names the numerator and comparator; red favors the first group, blue favors the second



D Randomized treatment effects are transient and program−dependent

Felzartamab−minus−placebo difference−in−differences; negative values indicate relative reduction



FDR < 0.05 FALSE · TRUE

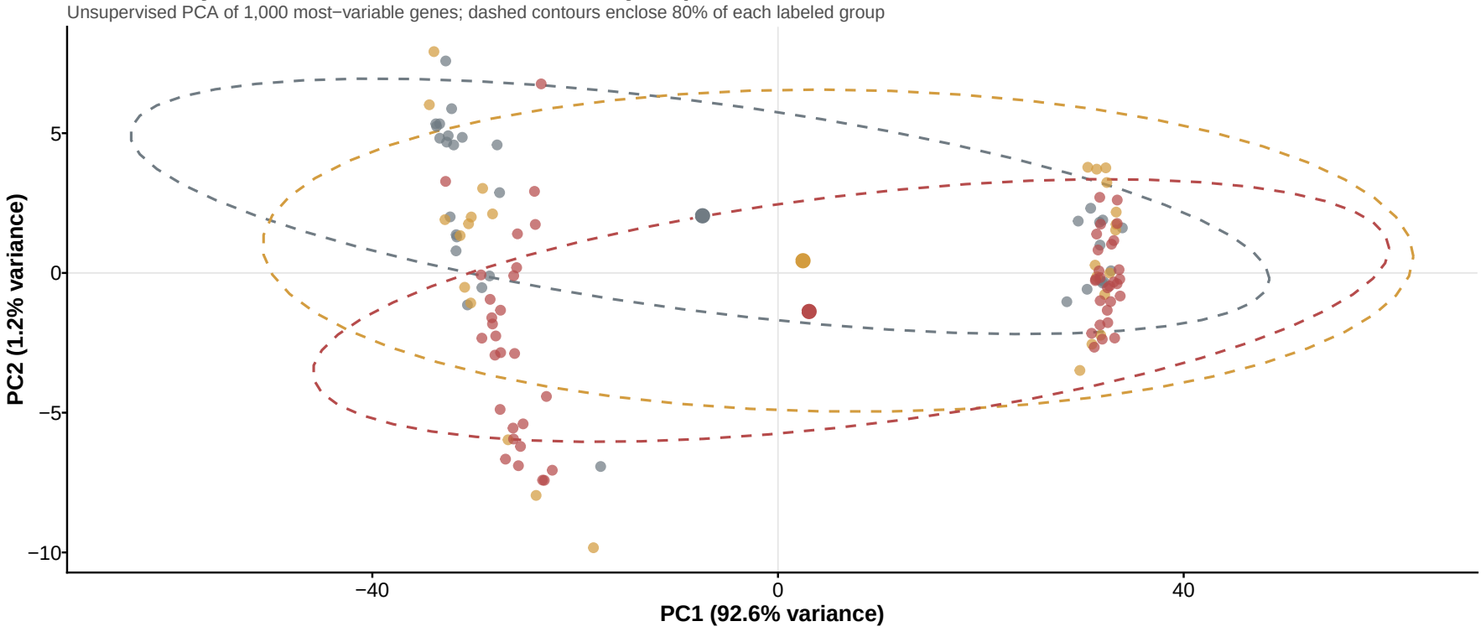
Hallmark NES −2 0 2 4

Time Week 24 · Week 52

Evidence FDR < 0.05 · FDR >= 0.05

E Discovery−cohort structure: GSE50084 kidney biopsies

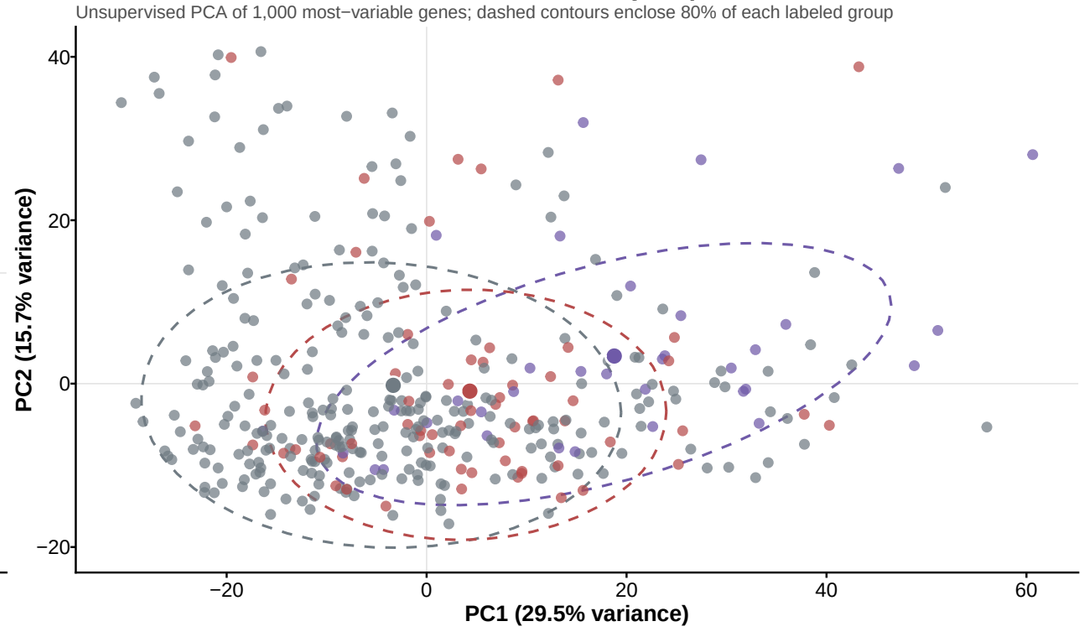
Unsupervised PCA of 1,000 most−variable genes; dashed contours enclose 80% of each labeled group



Clinical group DSA− / no rejection · DSA+ / no rejection · DSA+ / rejection

F Validation−cohort structure: GSE36059 kidney biopsies

Unsupervised PCA of 1,000 most−variable genes; dashed contours enclose 80% of each labeled group



Clinical group Non−rejecting · TCMR · ABMR

Associative bulk transcriptomics cannot prove antibody engagement, cell of origin, target causality, or drug efficacy.